

Environment Setup Cheat Sheet

Databases Project - Complete Installation Guide

1. PREREQUISITES

- **OS:** Windows with PowerShell
 - **Python:** 3.9+ (check with `python --version`)
 - **Git:** Installed and configured
-

2. MYSQL 8.4.9 INSTALLATION

Download

- Go to: <https://dev.mysql.com/downloads/mysql/>
- Select Version: **8.4.9**
- OS: **Windows 64-bit**
- Download: `mysql-8.4.9-winx64.msi`

Installation Steps

1. Double-click the `.msi` file
2. Choose: **Server only**
3. Port: Keep **3306**
4. Root Password: Set to `root` (or your preference)
5. Configure MySQL as Windows Service: ☒ **Check this**

Add to PATH (Windows)

```
powershell

# Add MySQL to current session
$env:Path += ";C:\Program Files\MySQL\MySQL Server 8.4\bin"

# Make it permanent
[Environment]::SetEnvironmentVariable("Path", $env:Path, "User")
```

Verify Installation

```
powershell

mysql --version
# Output: mysql Ver 8.4.9 for Win64 on x86_64
```

3. MONGODB 8.3.1 INSTALLATION

Download

- Go to: <https://www.mongodb.com/try/download/community>
- Select Version: **8.3.1** (or latest)
- OS: **Windows 64-bit**
- File: `mongodb-8.3.1-winx64.msi`

Installation Steps

1. Double-click the `.msi` file
2. Choose: **Custom Installation**
3. Install Location: Keep default
4. **Install MongoDB as a Service:** ☒ **Check this**
5. Service Name: Keep **MongoDB**

Add to PATH (Windows)

```
powershell

# Add MongoDB to current session
$env:Path += ";C:\Program Files\MongoDB\Server\8.3\bin"

# Make it permanent
[Environment]::SetEnvironmentVariable("Path", $env:Path, "User")
```

Verify Installation

```
powershell

mongod --version
# Output: db version v8.3.1
```

Start MongoDB Service

```
powershell  
  
net start MongoDB
```

4. PYMONGO INSTALLATION

Install via pip

```
powershell  
  
python -m pip install pymongo==4.6.2
```

Verify Installation

```
powershell  
  
python -c "import pymongo; print(pymongo.__version__)"  
# Output: 4.6.2
```

5. REPOSITORY SETUP

Clone Repository

```
powershell  
  
git clone https://github.com/YOUR_USERNAME/dlh-machine_learning.git  
cd dlh-machine_learning
```

Create Project Structure

```
powershell  
  
mkdir -p pipeline/databases  
cd pipeline/databases
```

Create README.md

```
markdown
```

Databases Project

Overview

This project covers **Relational Databases (MySQL)** and **Non-Relational Databases**

Requirements

- MySQL 8.4.9
- MongoDB 8.3.1
- Python 3.9+ with PyMongo 4.6.2
- pycodestyle 2.11.1

Tools

- MySQL: Relational database for structured data
- MongoDB: NoSQL database for flexible documents
- PyMongo: Python driver for MongoDB

6. VERIFY COMPLETE SETUP

```
powershell

# Check MySQL
mysql --version

# Check MongoDB
mongod --version

# Check Python
python --version

# Check PyMongo
python -c "import pymongo; print(pymongo.__version__)"

# Check Git
git --version
```

7. COMMON COMMANDS

MySQL Operations

```
powershell
```

```
# Connect to MySQL
mysql -uroot -p

# Create database
echo "CREATE DATABASE db_0;" | mysql -uroot -p

# Run SQL file
cat script.sql | mysql -uroot -p db_0

# Connect to specific database
mysql -uroot -p db_0
```

MongoDB Operations

```
powershell

# Start MongoDB
net start MongoDB

# Connect to MongoDB shell
mongo

# Stop MongoDB
net stop MongoDB
```

Git Workflow

```
powershell

# Check status
git status

# Add files
git add .

# Commit
git commit -m "Your message"

# Push
git push

# Pull
git pull
```

8. TROUBLESHOOTING

MySQL not recognized

```
powershell

# Re-add to PATH
$env:Path += ";C:\Program Files\MySQL\MySQL Server 8.4\bin"

# Or restart PowerShell
```

MongoDB not recognized

```
powershell

# Re-add to PATH
$env:Path += ";C:\Program Files\MongoDB\Server\8.3\bin"

# Or restart PowerShell
```

MySQL password issues

```
powershell

# Test connection
mysql -uroot -proot

# Reset password (if needed)
mysqladmin -uroot -p password "root"
```

Service won't start

```
powershell

# Check MySQL service
net start MySQL80

# Check MongoDB service
net start MongoDB
```

9. FILE STRUCTURE

```
dlh-machine_learning/
├─ README.md
```

```
|─ pipeline/
|   └─ databases/
|       └─ README.md
|       └─ 0-create_database_if_missing.sql
|       └─ 1-first_table.sql
|       └─ 2-list_values.sql
|       └─ 3-insert_value.sql
|       └─ 4-best_score.sql
|       └─ 5-average.sql
|       └─ 6-avg_temperatures.sql
|       └─ 7-max_state.sql
|       └─ setup.sql
└─ .git/
```

10. QUICK REFERENCE

Tool	Version	Command	Purpose
MySQL	8.4.9	<code>mysql --version</code>	Relational Database
MongoDB	8.3.1	<code>mongod --version</code>	NoSQL Database
Python	3.14.4	<code>python --version</code>	Programming Language
PyMongo	4.6.2	<code>python -c "import pymongo; print(pymongo.__version__)"</code>	MongoDB Driver
Git	Latest	<code>git --version</code>	Version Control

Last Updated: May 8, 2026 **Status:**  Complete Setup Ready